

SQL SELECT Practice Questions

- 25 Interview-Ready Questions

E-Commerce Database | Easy → Hard | All Important Topics

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✂ Quick Setup (Run First)

```
-- Run all CREATE and INSERT statements from
Capstone_Project_ECommerce_Database.md
-- Before attempting these questions
```

EASY - Basic SELECT (5 Questions)

Q1: Select All Users

Topic: Basic SELECT, SELECT *

Display all users with all their information.

- ▶ [Click to see Question Details](#)
 - ▶ [Click for Answer](#)
-

Q2: Select Specific Columns

Topic: Column Selection, Aliases

Display only the UserName and Email of all customers, rename Email column as "Contact Email".

- ▶ [Click for Answer](#)
-

Q3: Select All Products with Price

Topic: SELECT, WHERE with Simple Condition

Display all electronics products (products where category = 'Electronics').

- ▶ [Click for Answer](#)
-

Q4: Filter with Comparison Operator

Topic: WHERE with >, <, >=, <=

Find all products with price greater than 50000.

► [Click for Answer](#)

Q5: Simple Sorting

Topic: ORDER BY ASC/DESC

List all orders sorted by TotalAmount in descending order (highest first).

► [Click for Answer](#)

MEDIUM - JOINS & Filtering (10 Questions)

Q6: Simple INNER JOIN

Topic: INNER JOIN, Table Relationships

Display each order with the customer's name and city.

► [Click for Answer](#)

Q7: Multiple JOINS

Topic: Multiple JOINS, 3+ Table Join

Display each order item with: OrderID, Customer Name, Product Name, and Price.

► [Click for Answer](#)

Q8: LEFT JOIN to Find Missing Data

Topic: LEFT JOIN, Finding NULL values

Show all users and their orders. Include users who haven't placed any orders.

► [Click for Answer](#)

Q9: WHERE with AND/OR

Topic: Multiple Conditions, Logical Operators

Find all orders with status "Delivered" AND TotalAmount > 50000.

► [Click for Answer](#)

Q10: LIKE for Pattern Matching

****Topic: LIKE, Wildcards (% , _)****

Find all customers whose name starts with 'R' or contains 'Kumar'.

► [Click for Answer](#)

Q11: BETWEEN Range Query

Topic: BETWEEN Operator, Range Filtering

Find all products with price between 10000 and 50000.

► [Click for Answer](#)

Q12: IN Operator with Multiple Values

Topic: IN Operator, Multiple Values

Show all products in categories 'Electronics' or 'Fashion'.

► [Click for Answer](#)

Q13: NULL Checking

Topic: IS NULL, IS NOT NULL

Find all products with stock information (Stock is NOT NULL).

► [Click for Answer](#)

Q14: Calculated Columns

Topic: Expressions, Calculations, Aliases

For each product, show: ProductName, Price, and Price with 10% discount.

► [Click for Answer](#)

Q15: DISTINCT - Remove Duplicates

Topic: DISTINCT, Unique Values

Show all unique cities where customers are located.

► [Click for Answer](#)

HARD - Aggregates & GROUP BY (7 Questions)

Q16: COUNT Aggregate Function

Topic: COUNT, Aggregate Functions

Count total number of orders and total number of customers.

► [Click for Answer](#)

Q17: SUM - Calculate Totals

Topic: SUM, Aggregate Functions

Calculate total revenue from all orders.

► [Click for Answer](#)

Q18: GROUP BY - Aggregates Per Group

Topic: GROUP BY, GROUP BY with Aggregate

Show total revenue per customer.

► [Click for Answer](#)

Q19: GROUP BY with Multiple Columns

Topic: GROUP BY Multiple Columns, Aggregates

Show total orders and total revenue per order status.

► [Click for Answer](#)

Q20: HAVING - Filter Groups

Topic: HAVING Clause, Filtering Groups

Find customers who have spent more than 100,000 in total.

► [Click for Answer](#)

Q21: Top N Products by Sales

Topic: GROUP BY, ORDER BY, TOP, Aggregates

Find top 5 best-selling products by quantity.

► [Click for Answer](#)

Q22: Complex GROUP BY with JOINS

Topic: Multiple JOINS, GROUP BY, Multiple Aggregates

Show products sold per category with total quantity and revenue.

► [Click for Answer](#)

EXPERT - Complex & Real-World (3 Questions)

Q23: Subquery in WHERE Clause

Topic: Subqueries, Nested Queries

Find all orders with total amount greater than the average order value.

► [Click for Answer](#)

Q24: Subquery in SELECT (Derived Column)

Topic: Subqueries in SELECT, Correlation

For each customer, show their orders with percentage of total revenue.

► [Click for Answer](#)

Q25: Window Functions / Ranking

Topic: Window Functions (Advanced), ROW_NUMBER, RANK

Rank products by revenue within each category.

► [Click for Answer](#)

Solutions & Explanations

Summary of Key SQL Concepts Covered:

Topic	Questions	Key Points
Basic SELECT	Q1-Q5	SELECT *, columns, WHERE, ORDER BY
JOINS	Q6-Q8	INNER JOIN, Multiple JOINS, LEFT JOIN
WHERE Conditions	Q9-Q15	AND/OR, LIKE, BETWEEN, IN, NULL, Calculations
Aggregates	Q16-Q17	COUNT, SUM, AVG, MIN, MAX
GROUP BY	Q18-Q22	GROUP BY, HAVING, Multiple columns
Advanced	Q23-Q25	Subqueries, Window Functions

Interview Tips

Common Interview Questions About These Topics:

Q: Difference between WHERE and HAVING?

WHERE: Filters INDIVIDUAL ROWS before grouping
HAVING: Filters GROUPS after grouping

Q: Why use LEFT JOIN over INNER JOIN?

INNER JOIN: Only matching rows
LEFT JOIN: All left table rows + matching right rows
Use LEFT JOIN when you need to find missing/unmatched data

Q: What's wrong with this?

```
-- WRONG
SELECT UserID, UserName, COUNT(*)
FROM Orders
GROUP BY UserID; -- UserName not in GROUP BY!

-- CORRECT
SELECT UserID, UserName, COUNT(*)
FROM Users u
INNER JOIN Orders o ON u.UserID = o.UserID
GROUP BY u.UserID, u.UserName;
```

Q: Performance Tip

```
-- SLOW: Subquery in SELECT (runs for each row)
SELECT (SELECT COUNT(*) FROM Orders) AS Total FROM Users;

-- FASTER: Subquery once
SELECT COUNT(*) AS Total FROM Orders;
```

Interview Do's & Don'ts:

DO: - Always use table aliases for clarity - Use meaningful column names (AS 'alias') - Test WHERE clause with SELECT first before DELETE/UPDATE - Use proper JOIN conditions - Comment your complex queries

DON'T: - Use SELECT * in production code - Use = NULL or != NULL (use IS NULL) - Forget to include all non-aggregated columns in GROUP BY - Use INNER JOIN when LEFT JOIN is needed - Assume column order in SELECT matches ORDER BY position

Practice Tips

1. Start with Easy Questions

- Build confidence with basic SELECT
- Understand table structure first

2. Practice JOINS Extensively

- Most interview questions use JOINS
- Understand 1-to-Many and Many-to-Many relationships
- Practice LEFT JOIN for finding missing data

3. Master GROUP BY

- Very common in real-world queries
- Understand difference between WHERE and HAVING
- Practice with multiple columns

4. Understand Execution Order

FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

5. Test Your Queries

- Run each query against the E-Commerce database
 - Verify results make sense
 - Modify queries to understand changes
-

Next Steps

After mastering these 25 questions: 1. Try writing queries for business problems 2. Optimize queries for performance 3. Learn about indexes and query plans 4. Practice complex joins with 5+ tables 5. Learn about stored procedures and views

Difficulty Distribution

Difficulty	Count	Focus Area
Easy	5	Basics
Medium	10	JOINS & Filtering
Hard	7	Aggregates & GROUP BY
Expert	3	Advanced & Real-World
Total	25	Complete Coverage

Final Tips for Interviews

1. **Ask clarifying questions** before writing queries
 2. **Start simple** and build complexity
 3. **Use meaningful aliases** (u for Users, o for Orders)
 4. **Comment your code** - shows clear thinking
 5. **Test edge cases** - NULLs, empty results, large datasets
 6. **Performance matters** - Avoid nested loops, use JOINS efficiently
 7. **Practice writing** these queries by hand (without running them first)
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Good luck with your interview!

Practice these questions daily for 1 week and you'll master SQL SELECT!